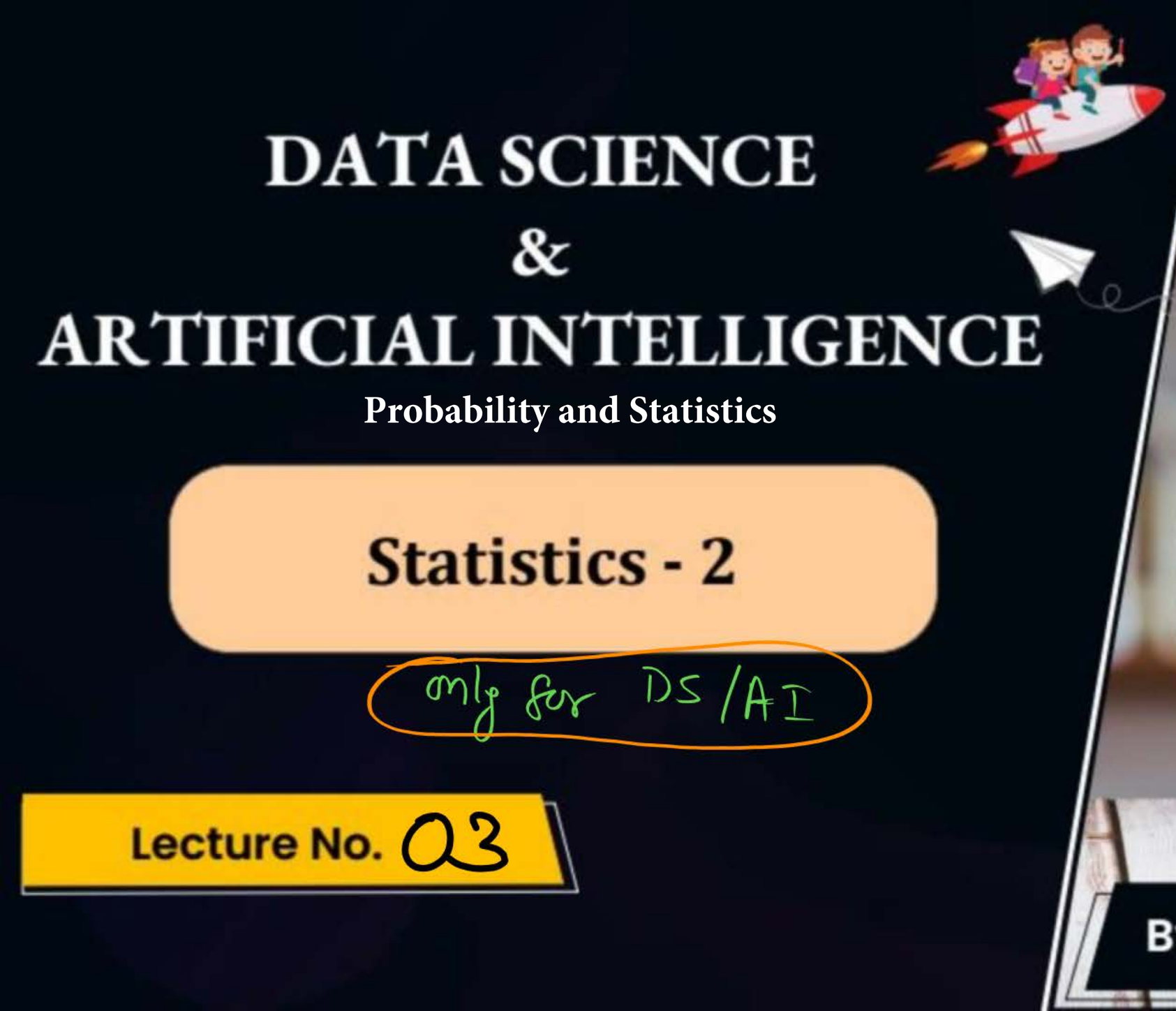
DATA SCIENCE
&
ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 2

only for DS/AI

Lecture No. 03



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

MULTIPLE INTEGRAL (BASICS)

Topics to be Covered



Topic

Application of Multiple Integral

(≡ useful for Joint p.d.f)

MULTIPLE INTEGRAL

Double Integral Working Rule :→

Case I: If y has variable limits, we should first integrate y keeping x constant.

Case II: if x " " " " " " x " y " "

Case III: if Both the variables have constant limits, we can integrate in any order, But it is advisable to integrate in that order which is given in Question.

If Both the limits are constants OR integrand is an Implicit func^n of x & y then we can integrate in any order keeping other variable constant.

Case IV-e If Both the limits are constants and Integrand is an Explicit func^n of x & y then we can integrate separately.

Tripple Integration Working Rule →

Case I: if all the variables have constant limits then we can integrate in any order keeping other variables constant

Case II: if one or two variables have variable limits then

"which one is more variable should be integrated 1st"

i.e.
$$I=\int_a^b\int_{f(x)}^{g(x)}\int_{\phi(x,y)}^{\psi(x,y)} f(x,y,z)\,dx\,dy\,dz = ? = \int_a^b\int_{f(x)}^{g(x)}\int_{\phi(x,y)}^{\psi(x,y)} f(x,y,z)\,dz\,dy\,dx$$

$x=$ $y=$ $z=$

Application of Multiple Integral →

① $\boxed{\text{Area} = \iint (1)\, dx\,dy} = \int y\,dx = \int_{x=a}^{b} f(x)\,dx$

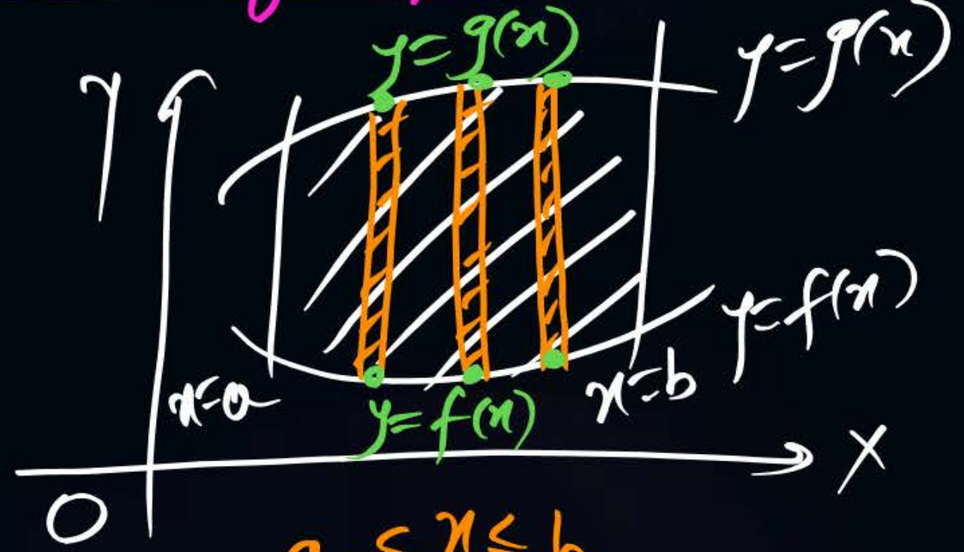


② $\boxed{\text{Volume} = \iiint (1)\, dx\,dy\,dz} = \iint z\,dx\,dy = \iint f(x,y)\,dx\,dy$

③ $\text{Mass} = \iiint (\rho)\, dx\,dy\,dz$

Change of order of Integration → If V. strip is given then for changing the order of Integration we should make H. strip & put the limits accordingly & Vice-Versa.

Nature of Strip :-

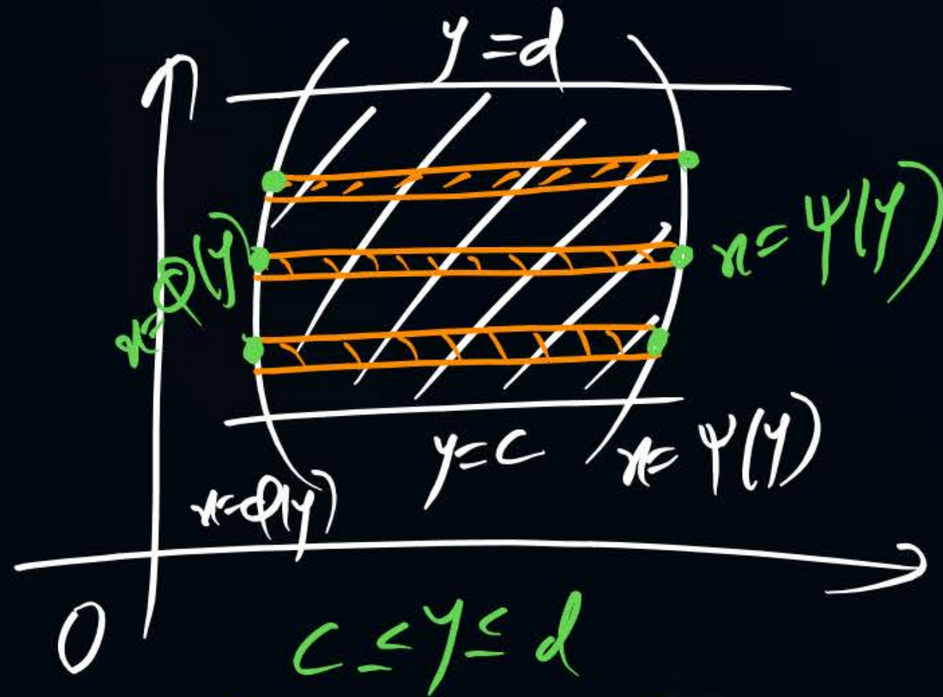


$a \le x \le b$

$f(x) \le y \le g(x)$

$$I = \int_{x=a}^{b} \int_{y=f(x)}^{g(x)} f(x,y)\,dy\,dx \neq \text{Area}$$

while $A = \int_{x=a}^{b} \int_{y=f(x)}^{g(x)} (1)\,\underline{dy\,dx}$ ✓



$c \le y \le d$

$\phi(y) \le x \le \psi(y)$

$$I = \int_{y=c}^{d} \int_{x=\phi(y)}^{\psi(y)} f(x,y)\,dx\,dy \neq \text{Area}$$

while $\text{Area} = \int_{y=c}^{d} \int_{x=\phi(y)}^{\psi(y)} (1)\,\underline{dx\,dy}$ ✓

Q: Find the area bounded by $y^2=x$ & $y=x$ using Both types of strips.

Sol: $y^2=x$ — (1) $\quad y=x$ — (2)

$x^2=x$

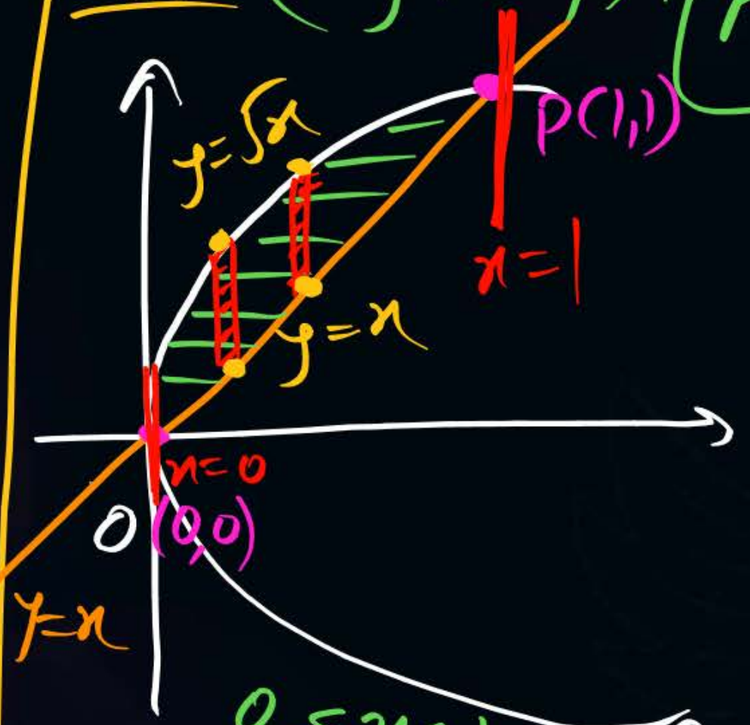
$x^2-x=0$

$x(x-1)=0$

$x=0$ & 1 so $y=0$ & 1

i.e. $O(0,0)$ & $P(1,1)$ are two Intersecting Points.

Case I (Using V-Strip) $\Rightarrow$ $A=\int\int (1)\,dy\,dx$



$0\le x\le 1$

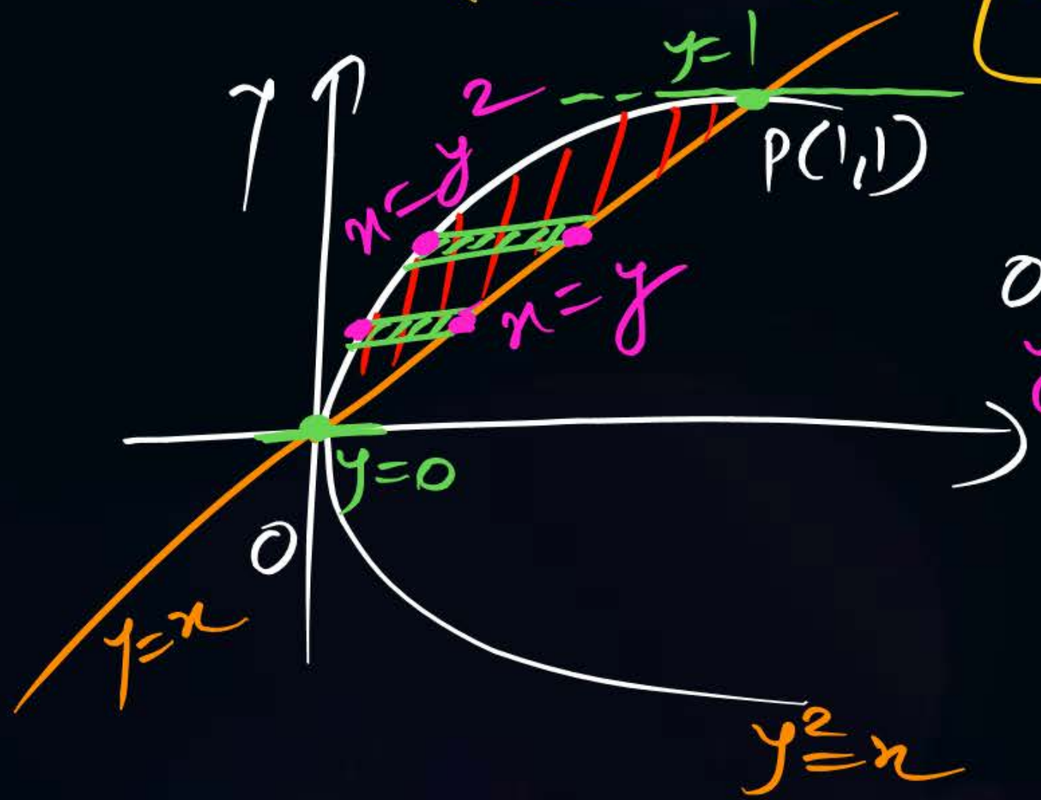
$x\le y\le \sqrt{x}$

$$=\int_{x=0}^{1}\int_{y=x}^{\sqrt{x}}(1)\,dy\,dx$$

$$=\int_{x=0}^{1}(\sqrt{x}-x)\,dx=\left(\frac{x^{3/2}}{3/2}-\frac{x^2}{2}\right)\Bigg|_0^1$$

$$=\ldots\ldots=\frac{1}{6}$$

Case II: (Using H-Strip) $\Rightarrow$ $\boxed{\text{Area} = \iint (1)\, dx\, dy}$ $= \int_{y=0}^{1} \int_{x=y^2}^{y} (1)\, dx\, dy$



$0 \le y \le 1$

$y^2 \le x \le y$

$= \int_{y=0}^{1} \{y - y^2\}\, dy$

$= \left(\frac{y^2}{2} - \frac{y^3}{3}\right)_{y=0}^{1} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$



Q: Find the intersecting points of $y^2 = x$ & $x^2 = y$?

Sol: $y^2 = x$ — (1), $x^2 = y$ — (2)

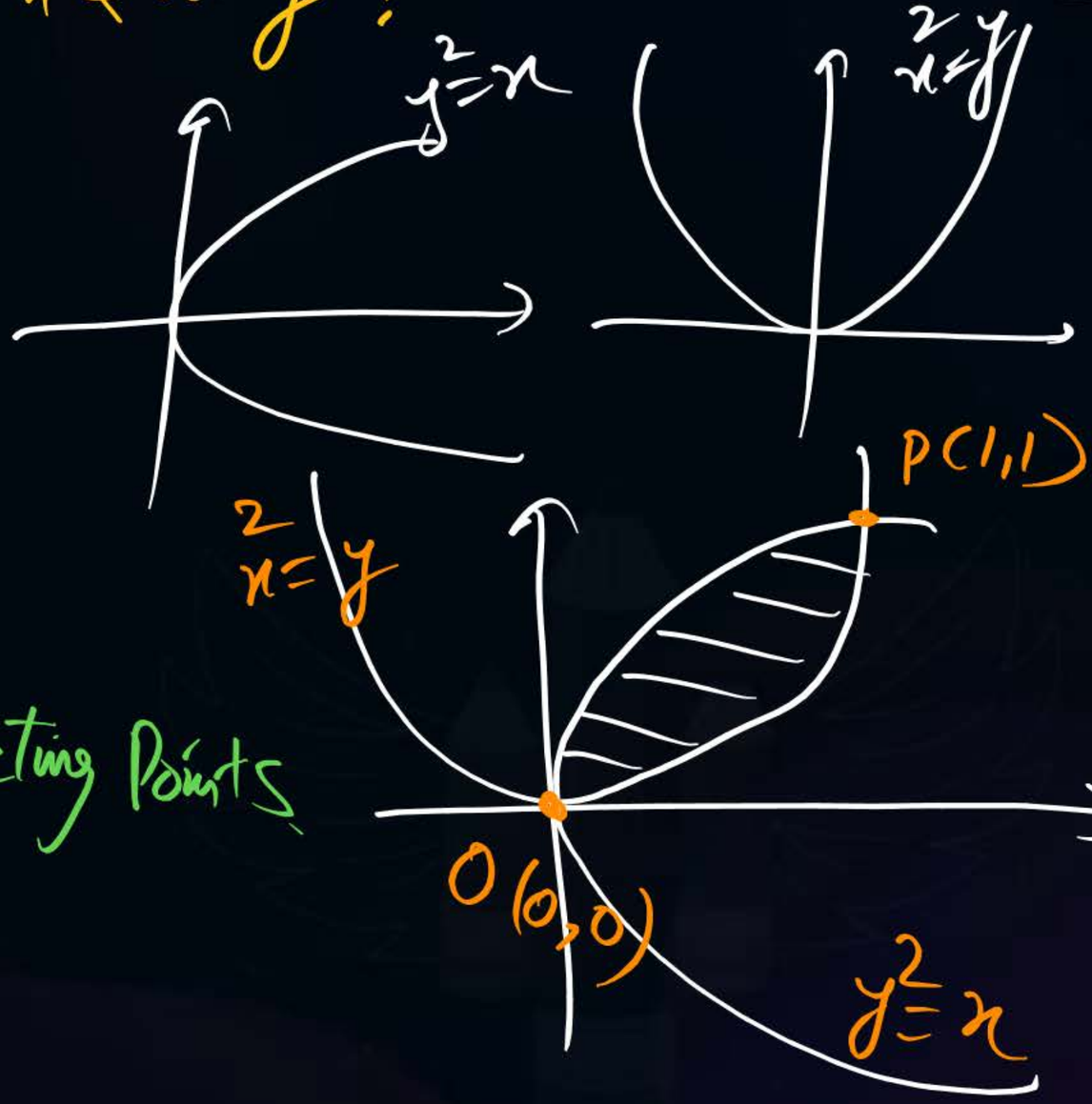
$(x^2)^2 = x$

$x^4 - x = 0$

$x(x^3 - 1) = 0$

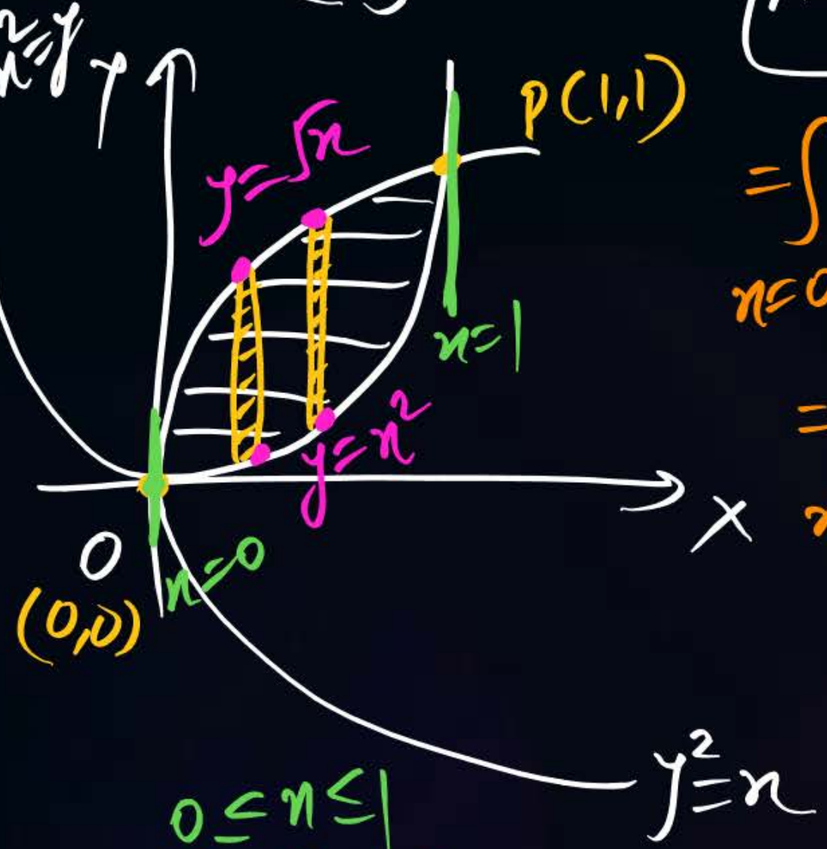
$x = 0$ & 1 then $y = 0$ & 1

is $O(0,0)$ & $P(1,1)$ are two intersecting Points.



Q. Find the area bounded by the parabolas $y^2 = x$ & $x^2 = y$ using both types of strips?

Case I: (Using V-Strip) → $\text{Area} = \iint (1)\, dy\, dx$



$$= \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} (1)\, dy\, dx$$

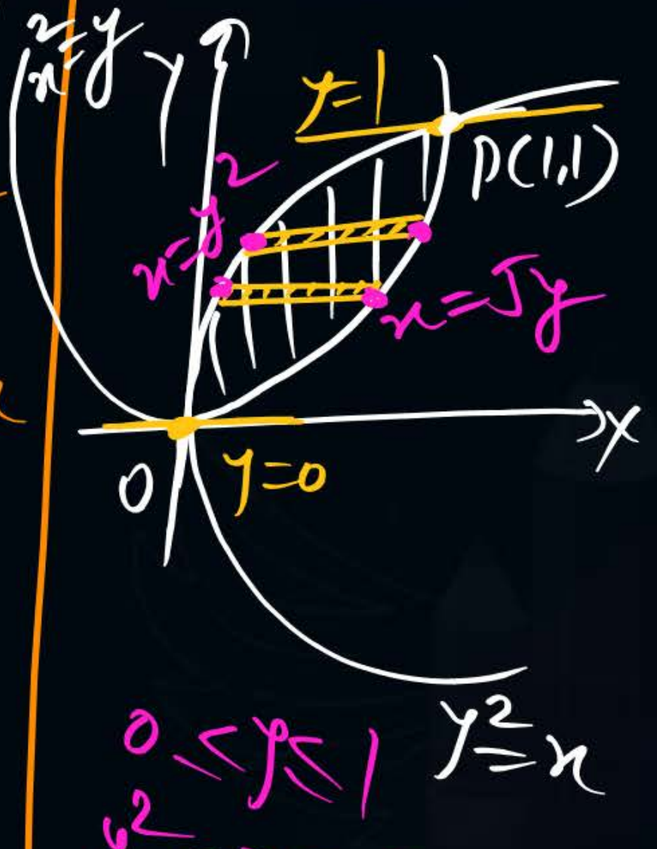
$$= \int_{x=0}^{1} (\sqrt{x} - x^2)\, dx$$

$$= \ldots = \frac{1}{3}$$

$0 \le x \le 1$

$x^2 \le y \le \sqrt{x}$

Case II (Using H-Strip) → $\text{Area} = \iint (1)\, dx\, dy$



$$= \int_{y=0}^{1} \int_{x=y^2}^{\sqrt{y}} (1)\, dx\, dy$$

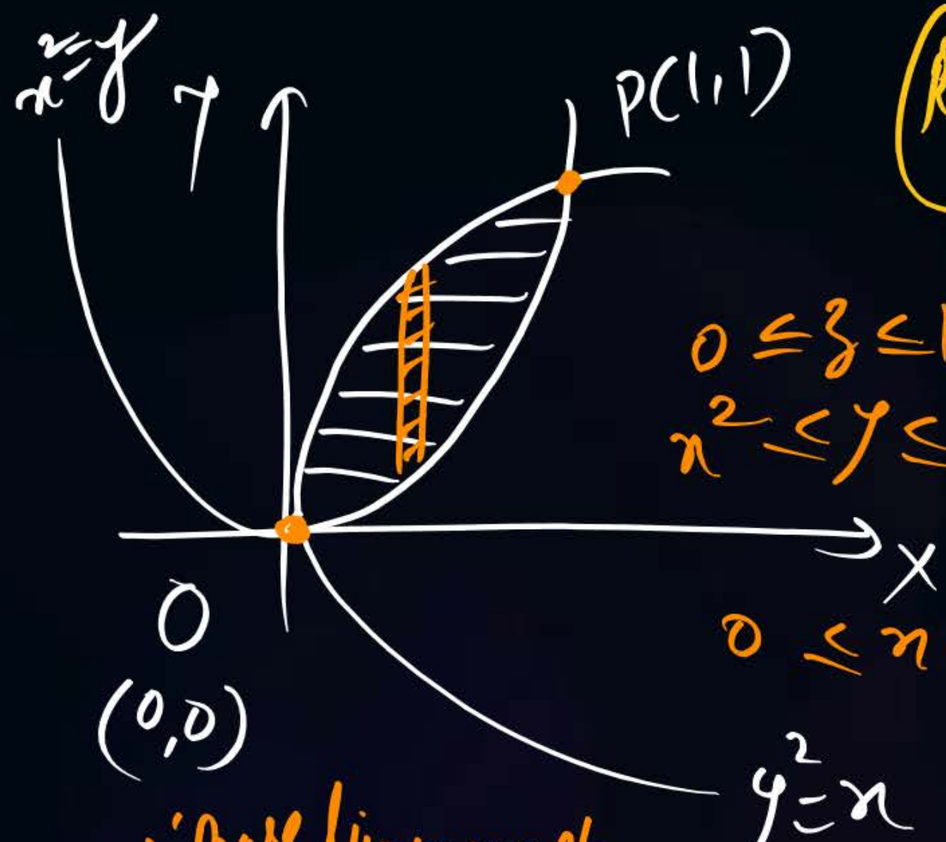
$$= \int_{y=0}^{1} (\sqrt{y} - y^2)\, dy$$

$$= \ldots = \frac{1}{3}$$

$0 \le y \le 1$

$y^2 \le x \le \sqrt{y}$

Q. Find the Volume of solid, standing on the area common to the parabolas $y^2=x$ & $x^2=y$ (as Base) and cut by the surface $z=12+y-x^2$?



$\therefore$ Base lies on xy plane then $z=0$

$0 \le z \le 12+y-x^2$

$x^2 \le y \le \sqrt{x}$

$0 \le x \le 1$

Req Volume $= \iiint (1)\, dx\,dz\,dy\,dz$

$$= \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} \int_{z=0}^{12+y-x^2} (1)\, dz\, dy\, dx$$

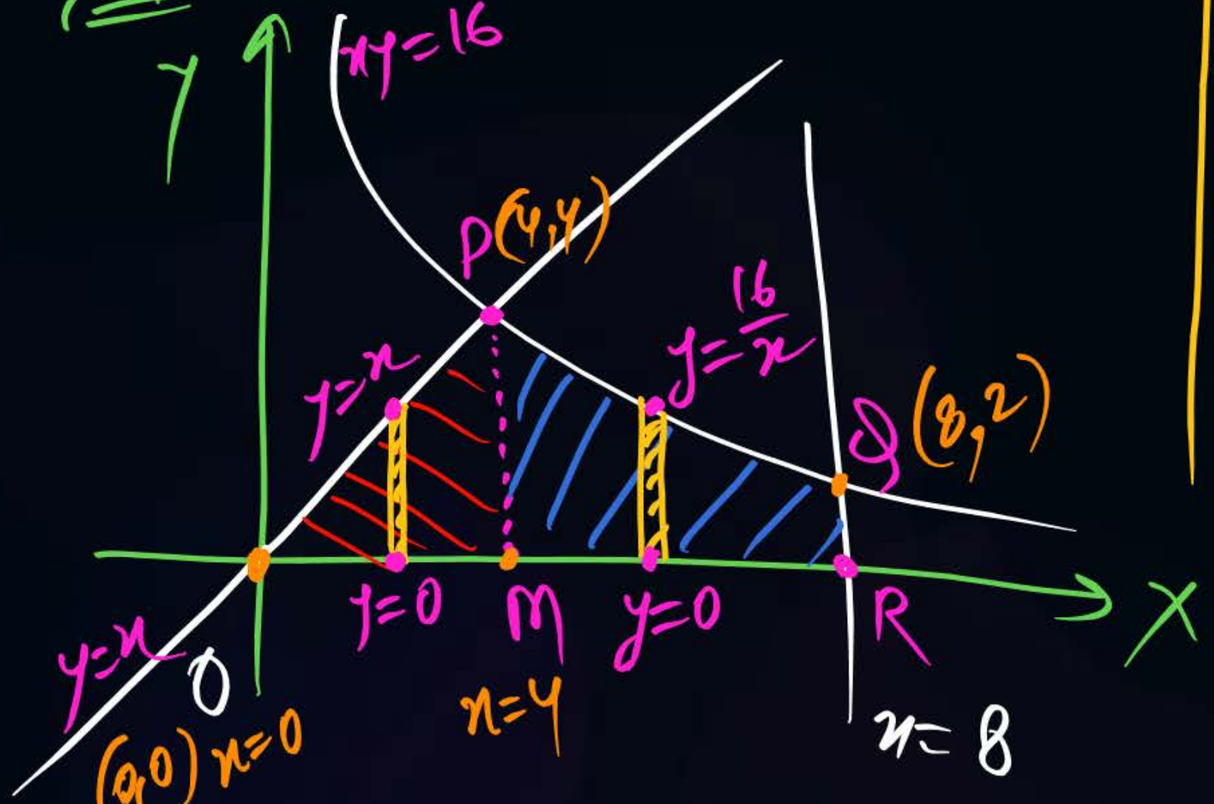
$$= \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} (12+y-x^2)\, dy\, dx$$

$$= \int_{x=0}^{1} \left\{ 12y + \frac{y^2}{2} - x^2 y \right\}_{y=x^2}^{\sqrt{x}} \cdot (dx) = \cdots = \frac{569}{140}$$

Q: Evaluate $\iint_D x^2\,dx\,dy = ?$ where D is the region in 1st Quadrant bounded by

$xy = 16$, $y = 0$ (x axis), $y = x$, $x = 8$ (V. line)

Sol:



Req. Region of Integration is OPMRQ

In OPM; $0 \le x \le 4$, $0 \le y \le x$

In MPQR; $4 \le x \le 8$, $0 \le y \le \frac{16}{x}$

$$I = \iint_D x^2\,dy\,dx = \iint_{OPM} + \iint_{MPQR}$$

$$I = \int_{x=0}^{4}\int_{y=0}^{x} (x^2)\,dy\,dx + \int_{x=4}^{8}\int_{y=0}^{16/x} x^2\,dy\,dx$$

$$= \int_{x=0}^{4}\left\{\int_{y=0}^{x}(1)\,dy\right\}(x^2\,dx) + \int_{x=4}^{8}\left\{\int_{y=0}^{16/x}(1)\,dy\right\}(x^2\,dx)$$

$$= \int_{x=0}^{4}(x-0).(x^2\,dx) + \int_{x=4}^{8}\left(\frac{16}{x}-0\right).(x^2)\,dx$$

$$= \int_{0}^{4} x^3\,dx + \int_{x=4}^{8}(16x)\,dx = \left(\frac{x^4}{4}\right)_0^4 + (8x^2)_4^8 = \ldots = 448$$

it is Not the area of above Region $\because$ Integrand is not of 1 unit.

Q: find the area bounded by $xy=16$, $y=0$, $y=x$, $x=8$?

Using previous Diagram,

$$\text{Area} = \int\int (1)\,dy\,dx = \int_{x=0}^{4}\int_{y=0}^{x} (1)\,dy\,dx + \int_{x=4}^{8}\int_{y=0}^{16/x} (1)\,dy\,dx$$

$$= \int_{x=0}^{4} (x)\,dx + \int_{x=4}^{8} \left(\frac{16}{x}\right)dx$$

$$= \left(\frac{x^2}{2}\right)_0^4 + \left(16\ln x\right)_4^8$$

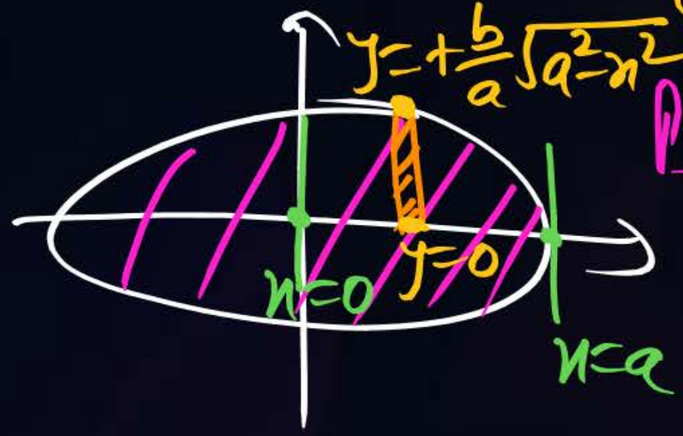
$$= 8 + 16\ln 2 \quad \underline{\text{Ans}}$$

Note-e (1) Area bounded by $y^2 = 4ax$ & $x^2 = 4by$ is $\frac{16}{3}ab$

eg " " " $y^2 = 4x$ & $x^2 = 4y$ is $\frac{16}{3}$ ($\because a=1$ & $b=1$)

eg " " " $y^2 = x$ & $x^2 = y$ $\frac{1}{3}$ ($\because 4a=1$ & $4b=1$)

(2) Area bounded by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab HW



Proof: Req Area $= 4$ Area in 1st Quadrant

$$= 4\int\int (1)\,dy\,dx = 4\int_{x=0}^{a}\int_{y=0}^{\frac{b}{a}\sqrt{a^2-x^2}} (1)\,dy\,dx = \cdots = \pi ab$$

Note:-

(1) $\log a + \log b = \log ab$

(2) $\log a - \log b = \log\left(\frac{a}{b}\right)$

(3) $\log a^b = b \log a$

(4) $\log_a x = b \iff x = a^b$

(5) $\log_{a^b} x = \frac{1}{b} \log_a x$

(6) $\log_y x = \frac{\log_a x}{\log_a y}$

(7) $\log_y x = \frac{1}{\log_x y}$

$\left\{ \because \log_y x = \frac{\log_x x}{\log_x y} = \frac{1}{\log_x y} \text{ Proved} \right\}$

(8) $\boxed{\log_a a = 1}$ (9) $\boxed{\log_a 1 = 0}$

(10) $\log_e(\infty) = +\infty$ i.e N.D

(11) $\log_e(0) = -\infty$ i.e N.D

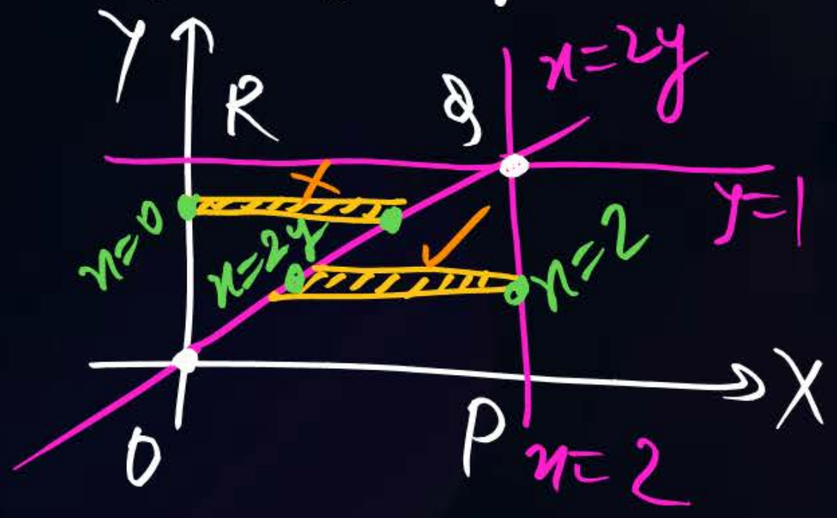
Q. $I = \int_0^1 \int_{2y}^{2} e^{x^2} \, dx \, dy = ?$

$= \int_{y=0}^{1} \left\{ \int_{x=2y}^{2} e^{x^2} \, dx \right\} dy = \int_{y=0}^{1} \{ \text{Not possible to integrate} \} \, dy$

is using this step, it is not possible to solve our question.

Then we will use the concept of change of order of integration.

Region of integration is bounded by $x = 2y$, $x = 2$, $y = 0$, $y = 1$ (i.e. OPQ)





x axis

$0 \le y \le \frac{x}{2}$

$0 \le x \le 2$

$$I=\int_0^1\int_{2y}^{2} e^{x^2}\,\frac{dx\,dy}{\text{H.Strip}}=\int_{x=0}^{2}\int_{y=0}^{x/2} e^{x^2}\,\frac{dy\,dx}{\text{V.Strip}}=\int_{x=0}^{2}\left\{\int_{y=0}^{x/2}(1)\,dy\right\}(e^{x^2}dx)$$

$$=\int_{x=0}^{2}\left\{\frac{x}{2}-0\right\}e^{x^2}dx=\int_{x=0}^{2}\frac{x}{2}e^{x^2}dx$$

Put $x^2=t$

$x\,dx=\frac{dt}{2}$

At $x=0$, $t=0$

At $x=2$, $t=4$

$$=\int_{t=0}^{4}\frac{1}{2}e^{t}\cdot\frac{dt}{2}=\frac{1}{4}\int_0^4 e^t\,dt$$

$$=\frac{1}{4}\left(e^4-e^0\right)=\frac{e^4-1}{4}$$

Let $\int_0^1 \int_y^1 xy \sin(xy)\,dx\,dy = \int_0^1 \int_a^b xy \sin xy\,dy\,dx$

H. Strip

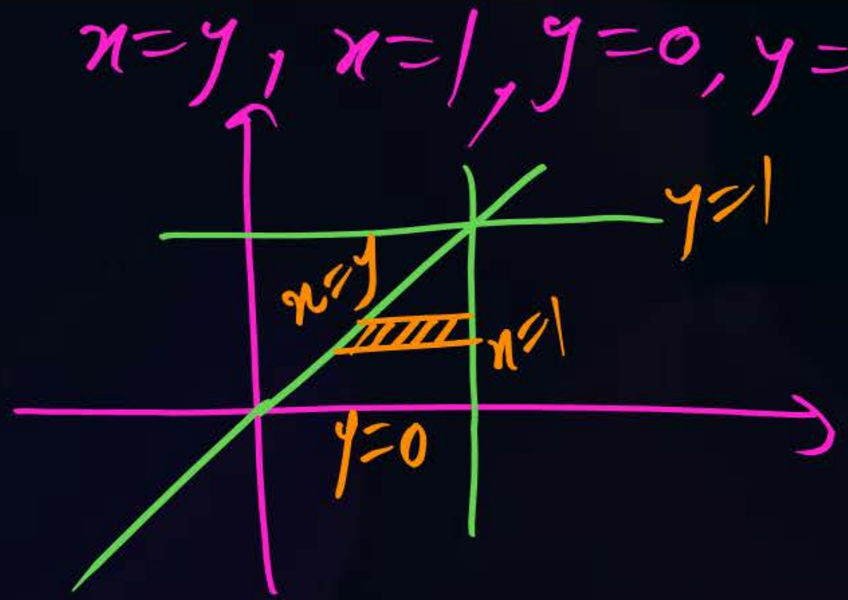
V. Strip

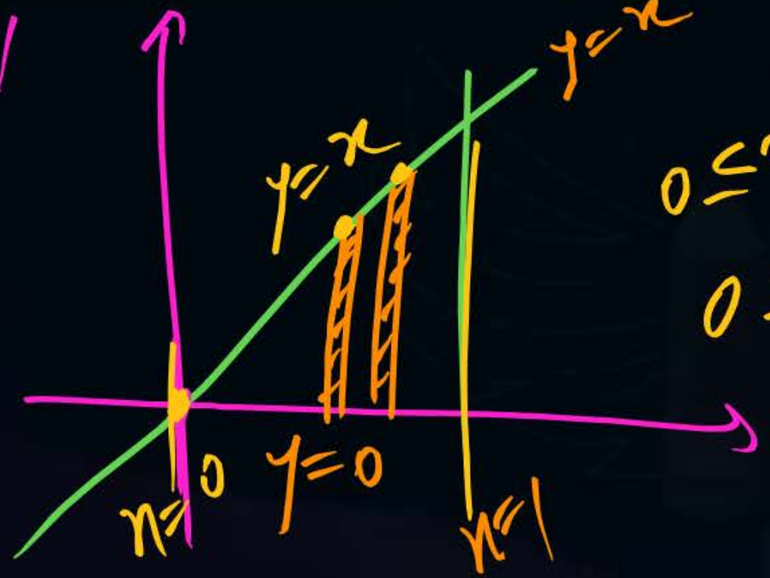
(a) $a = 0, b = x$ (b) $a = 1, b = x$

(c) $a = 0, b = 1$ (d) $a = -1, b = x$

$$I = \int_0^1 \int_y^1 f(x,y)\,dx\,dy$$

$$= \int_{x=0}^{1} \int_{y=0}^{x} f(x,y)\,dy\,dx \Rightarrow a = 0, b = x$$





$0 \le x \le 1$

$0 \le y \le x$

Q. $I = \int_0^\infty \int_0^\infty e^{-xy} \sin px \, dy \, dx = ? = \int_{x=0}^{\infty} \int_{y=0}^{\infty} e^{-xy} \sin px \, dy \, dx$ → M-I, M-II

M-I (Using V-Strip) →

$$I = \int_{x=0}^{\infty} \left\{ \int_{y=0}^{\infty} e^{-xy} \, dy \right\} (\sin px \, dx) = \int_{x=0}^{\infty} \left\{ \frac{e^{-xy}}{-x} \right\}_{y=0}^{\infty} (\sin px \, dx)$$

$$= \int_{x=0}^{\infty} \left\{ -\frac{1}{x} (e^{-\infty} - e^{0}) \right\} (\sin px \, dx) = \int_{x=0}^{\infty} \frac{\sin px}{x} \, dx = \int_0^\infty (??) \, dx$$

i.e. Using V-Strip, it is not possible to integrate

Hence we will use the Concept of change of order of integration.

M-II (Using H.Strip) →

$$I = \int_{x=0}^{\infty}\int_{y=0}^{\infty} e^{-xy}\sin px \, dy\,dx = \int_{y=0}^{\infty}\int_{x=0}^{\infty} e^{-xy}\sin px \, dx\,dy$$

$$= \int_{y=0}^{\infty}\left\{\int_{x=0}^{\infty} e^{-xy}\sin px \, dx\right\}(dy) = \text{- - - - - - - - - - HW - - - - - - - - - -} = \frac{\pi}{2}$$

use $\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}\left(a\sin bx - b\cos bx\right)$

joint PDF: (TEASER) or POSTER.

If x & y are Random Variable & $f(x,y)$ is it's joint p.d.f then

$$\Rightarrow P(a<x<b,\ c<y<d) = ? = \int_{y=c}^{d}\int_{x=a}^{b} f(x,y)\,dx\,dy$$

$$= ??$$

THANK - YOU